

Chapter 1 Assessment: Data Mining — Foundations & Key Tools

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Student Information

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Academic Integrity Statement

By submitting this assessment, I affirm the following: I have completed this assessment independently, using only permitted course materials (lecture slides, textbook, and my own notes). I have not copied answers from another student, an online solution set, or an AI tool, and I have not given my answers to another student. I understand that submitting work that is not my own is a violation of academic integrity policy and may result in disciplinary action.

Signature (type full name):

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I acknowledge and agree to the statement above.

Instructions

- Answer all questions in the space provided.
- Section D requires you to show your calculation steps, not just a final number.
- Where a question asks you to justify your answer, a one-line reason is sufficient unless stated otherwise.
- Estimated time: 45–60 minutes.

Section A: Multiple Choice (1 mark each)

- A1. Which of the following best describes "data mining" in the general (non-ML) sense used in this chapter?
- (a) Always training a machine learning classifier
 - (b) Building an algorithm to summarize or extract value from data (e.g., LSH, stream-mining)
 - (c) Only applies to numerical data
 - (d) A synonym for database management
- A2. In statistical modeling, what is the primary goal of fitting a distribution to data?
- (a) To increase the size of the dataset
 - (b) To assume an underlying distribution from which the visible data was drawn, and understand the problem better
 - (c) To remove the need for any further analysis
 - (d) To guarantee 100% prediction accuracy
- A3. Machine learning is described as uncompetitive when:
- (a) We have little idea what the data says about the problem
 - (b) We don't care about explainability
 - (c) We can describe the mining goal directly, or need interpretable reasoning
 - (d) The dataset is very large
- A4. What does PageRank fundamentally represent?
- (a) The number of words on a page
 - (b) The probability that a random walker on the web graph is at that page at any given time
 - (c) The page's file size
 - (d) The number of times a page has been edited
- A5. According to the Bonferroni Principle, what happens if you test too many hypotheses on random data?
- (a) You will find only true, meaningful patterns
 - (b) Some patterns will look significant purely by chance, even though they're statistical artifacts
 - (c) The dataset will become biased
 - (d) The algorithm will fail to run

Section B: True / False (1 mark each — briefly justify each answer)

B1. Computational approaches to modeling always aim to fully understand the process that generated the data.

True False

Justification:

B2. In TF-IDF, a term that appears in very few documents overall, but frequently within one document, tends to get a high TF-IDF score.

True False

Justification:

B3. A hash table can be used as a simple way to build an index.

True False

Justification:

B4. Power laws appear as a straight line when plotted on a standard (non-log) linear scale.

True False

Justification:

B5. Reading a single byte from a disk block costs roughly the same as reading the entire block, so algorithms should minimize the number of blocks touched.

True False

Justification:

Section C: Short Answer (2–3 marks each)

C1. Explain, in your own words, the difference between the statistical modeling approach and the computational modeling approach to data mining. When would you prefer one over the other?

C2. Describe how clustering achieves "summarization" of a dataset, using your own example (not the one from the slides).

C3. Explain why "finding people's resumes on the web" is given as an example where machine learning is a poor fit, while computer vision tasks are a good fit.

C4. In your own words, describe what an index is and why a hash table is a natural way to implement one.

Section D: Problem Solving (show your work)

D1. Fitting a Gaussian

Given the dataset: {3, 5, 5, 5, 6, 6, 8, 10}

(a) Compute the mean.

(b) Compute the variance and standard deviation.

(c) A new observation comes in at $x = 14$. How many standard deviations away from the mean is it? Would you consider it unusual? Justify briefly.

D2. TF-IDF Calculation

Given three documents — D1: "mining data mining tools" | D2: "data tools rock" | D3: "mining gold ore"

(a) Compute TF for "mining" in D1 (normalized by the max term frequency in D1).

(b) Compute IDF for "mining" given $N = 3$ documents.

(c) Compute the final TF-IDF score for "mining" in D1.

D3. Bonferroni Principle

2,000 people, 50 hotels, tracked over 20 days. $P(\text{two specific people, same hotel, same day}) = 1/50$.

(a) Expected same-hotel-same-day coincidences for one pair over 20 days. (b) Using Poisson approx ($P(X \geq 2) = 1 - e^{-\lambda}(1 + \lambda)$), estimate $P(\text{at least 2 coincidences})$ for that pair. (c) With 1,999,000 total pairs, estimate the expected number of "suspicious" pairs.

D4. Hash Function Bucketing

Using $h(k) = k \bmod 8$, compute the bucket for each key: 15, 23, 31, 40, 47.

(a) List the bucket assigned to each key. (b) Identify any collisions.

D5. The Constant e

(a) Using $(1+a)^b$ is approx $e^{(ab)}$ for small a , estimate $(1.02)^{40}$.

(b) Compute the exact value of $(1.02)^{40}$ (to 3 decimal places) and compare it to your estimate.

Section E: Applied / Design Thinking (open-ended, 3–5 marks each)

E1. A news website wants to detect near-duplicate articles being republished across the web (feature extraction / similar items). Describe, at a high level, how you would approach this problem, and which subtopic from this chapter (frequent itemsets, similar items, clustering, etc.) is most relevant.

E2. Explain why the Bonferroni Principle is especially important in the era of "big data," where analysts can run thousands of automated queries on a dataset. Give a realistic scenario (not from the slides) where ignoring this principle could lead to a false conclusion.

E3. A colleague says: "Since machine learning models can approximate almost any function, we should always prefer ML over statistical modeling or hand-built computational algorithms." Do you agree or disagree? Justify your answer using at least two concepts from this chapter (e.g., explainability, data availability, problem specificity).

End of Assessment — please review your Academic Integrity acknowledgment on page 1 before submitting.